

A Unified DTQEM Framework: Phonon Occupation, Zero-Point Fluctuations, and Multi-Phonon Relaxation in NV Centers

Reddouane BERRAMDANE

Independent Researcher

reddoma@gmail.com

June 2026

Abstract

Building on Paper I, where a single effective phonon mode with wavenumber $\tilde{\nu} = 711 \pm 48 \text{ cm}^{-1}$ provided an accurate effective description of the temperature-dependent zero-field splitting $D(T)$ of NV centers, we extend the DTQEM framework to spin-lattice relaxation $T_1(T)$ and propose a corresponding fluctuation ansatz for pure dephasing. We demonstrate that the fixed-frequency single-mode Raman factor $n(\omega, T)[n(\omega, T) + 1]$ is mathematically bounded by a T^2 high-temperature asymptote and cannot reproduce the T^5 -like Raman scaling observed experimentally over intermediate temperature ranges in NV centers. Replacing the discrete mode with an effective Debye–Raman transport integral $J_5(\Theta_{\text{eff}}/T)$ provides a physically interpretable continuum description of this scaling and yields a consistent effective cutoff $\Theta_{\text{eff}} = 596 \pm 88 \text{ K}$ across three Jarmola ensemble samples. The identity $\coth(x) = 2n(x) + 1$ provides a unified thermal language linking static shifts, fluctuation-driven dephasing ansatzes, and multi-phonon relaxation within a single phonon-occupation framework.

1 Introduction

The nitrogen-vacancy (NV) center in diamond is a leading platform for quantum sensing, with applications ranging from nanoscale thermometry to single-spin magnetic resonance [2, 3]. The performance of NV-based sensors is fundamentally limited by spin relaxation and decoherence, both of which are strongly temperature-dependent [4, 5]. The temperature dependence of the zero-field splitting and its relation to phonons, thermal expansion, and spin-phonon coupling have been discussed in several theoretical and experimental works [6, 7].

In Paper I of this series [1], we introduced the Decoherence-Temperature Quantum Effective Model (DTQEM) and showed that a single effective phonon occupation model accurately captures $D(T)$. Within that effective description, $D(T)$ primarily reflects a static, thermally averaged lattice-displacement scale. The longitudinal spin relaxation time $T_1(T)$ and the pure dephasing rate $\Gamma_\phi(T)$ are different observables. They are governed by dynamical fluctuations of the phonon bath and are often described by separate

empirical or semi-empirical laws. Here we provide a physically motivated effective framework that organizes $D(T)$, quantitatively models $T_1(T)$, and suggests a testable ansatz for $\Gamma_\phi(T)$ within a common phonon-occupation language.

The central statistical identity is

$$\coth(x) = \frac{2}{e^{2x} - 1} + 1 = 2n(\tilde{\nu}, T) + 1, \quad x = \frac{hc\tilde{\nu}}{2k_B T}, \quad (1)$$

where

$$n(\tilde{\nu}, T) = \frac{1}{\exp(hc\tilde{\nu}/k_B T) - 1}. \quad (2)$$

Thus the same Bose–Einstein statistics may appear as n , $2n+1$, or $n(n+1)$ depending on whether the observable probes static shifts, fluctuations, or Raman absorption-emission processes.

2 Theoretical Framework

2.1 Effective spin-phonon Hamiltonian

The NV ground-state spin is an effective $S = 1$ system. Since the zero-field splitting D_0 is conventionally expressed in frequency units, the corresponding energy Hamiltonian is written as

$$\hat{H} = hD_0\hat{S}_z^2 + \hbar\omega a^\dagger a + \lambda(a + a^\dagger)^2\hat{S}_z^2. \quad (3)$$

Here a and $a^\dagger$ are phonon annihilation and creation operators, $\omega = 2\pi c\tilde{\nu}$, and λ is an effective quadratic spin-strain coupling energy. This model retains the leading quadratic sensitivity of the zero-field splitting to local strain. Macroscopic thermal expansion, anharmonicity, and orbital electron-phonon effects are absorbed phenomenologically into the fitted parameters.

2.2 From $n(\tilde{\nu}, T)$ to $\coth(\tilde{\nu}, T)$

For an effective phonon mode of wavenumber $\tilde{\nu}$, the Bose–Einstein occupation number is

$$n(\tilde{\nu}, T) = \frac{1}{\exp(hc\tilde{\nu}/k_B T) - 1}. \quad (4)$$

If

$$\hat{u} = u_0(a + a^\dagger), \quad u_0 = \sqrt{\frac{\hbar}{2M\omega}}, \quad (5)$$

then the thermal mean-square displacement is

$$\langle \hat{u}^2 \rangle_T = u_0^2[2n(\tilde{\nu}, T) + 1] = u_0^2 \coth\left(\frac{hc\tilde{\nu}}{2k_B T}\right). \quad (6)$$

Since the zero-point contribution is temperature-independent, it is absorbed into the redefined constant D'_0 . For the static zero-field splitting shift we thus write

$$D(T) = D'_0 + A n(\tilde{\nu}, T). \quad (7)$$

This is the single-mode occupation model used in Paper I. For pure dephasing, the same occupation language suggests the effective fluctuation ansatz

$$\Gamma_\phi(T) = \Gamma_{\text{bg}} + \alpha \coth\left(\frac{\hbar c \tilde{\nu}_\phi}{2k_B T}\right). \quad (8)$$

Here Γ_{bg} summarizes non-phononic background noise, including impurities, the ^{13}C nuclear spin bath, strain, electric-field noise, and technical noise, while α is an effective phonon-mediated dephasing amplitude.

2.3 Origin and T^2 ceiling of the single-mode Raman rate

A two-phonon Raman process can arise either from second-order perturbation theory in a linear spin-phonon coupling or from an effective quadratic bath operator

$$\hat{B} \propto (a + a^\dagger)^2. \quad (9)$$

In a single-mode Raman approximation, the absorption-emission thermal factor has the form

$$\Gamma_R^{\text{single}}(T) \propto n(\omega, T)[n(\omega, T) + 1]. \quad (10)$$

Let

$$x = \frac{\hbar\omega}{k_B T}. \quad (11)$$

For $x \ll 1$,

$$n(x) = \frac{1}{x} - \frac{1}{2} + \frac{x}{12} + \mathcal{O}(x^3), \quad (12)$$

and therefore

$$n(n+1) = \frac{e^x}{(e^x - 1)^2} = \frac{1}{x^2} - \frac{1}{12} + \mathcal{O}(x^2). \quad (13)$$

The leading high-temperature algebraic growth is therefore T^2 . No higher positive power of T can be generated within a fixed-frequency single-mode Raman ansatz with temperature-independent coupling.

2.4 Effective Debye–Raman transport integral

A more realistic Raman process involves pairs of phonons drawn from a continuum. Microscopically,

$$\Gamma_R \propto \int d\omega J(\omega)J(\omega + \omega_s)n(\omega)[n(\omega + \omega_s) + 1]. \quad (14)$$

For NV centers, ω_s lies in the GHz range, while thermally relevant lattice modes are typically in the THz range. Therefore $\omega_s \ll \omega$, and one may approximate

$$J(\omega + \omega_s) \simeq J(\omega), \quad n(\omega + \omega_s) \simeq n(\omega). \quad (15)$$

This motivates the effective form

$$\Gamma_R(T) \propto \int_0^{\omega_c} d\omega W(\omega)n(\omega, T)[n(\omega, T) + 1], \quad (16)$$

where $W(\omega)$ is an effective two-phonon spectral weight. In the minimal Debye–Raman model we take

$$W(\omega) \propto \omega^4. \quad (17)$$

Introducing

$$x = \frac{\hbar\omega}{k_{\text{B}}T}, \quad \Theta_{\text{eff}} = \frac{\hbar\omega_c}{k_{\text{B}}}, \quad (18)$$

one obtains

$$\Gamma_R(T) \propto T^5 \int_0^{\Theta_{\text{eff}}/T} x^4 \frac{e^x}{(e^x - 1)^2} dx. \quad (19)$$

We define

$$\boxed{J_5(y) = \int_0^y x^4 \frac{e^x}{(e^x - 1)^2} dx}, \quad y = \frac{\Theta_{\text{eff}}}{T}. \quad (20)$$

The corresponding relaxation model is

$$\boxed{\frac{1}{T_1(T)} = \Gamma_0 + R \left(\frac{T}{\Theta_{\text{eff}}} \right)^5 J_5 \left(\frac{\Theta_{\text{eff}}}{T} \right)}. \quad (21)$$

The limiting forms are

$$J_5(y) \xrightarrow{y \rightarrow \infty} \frac{4\pi^4}{15}, \quad J_5(y) \xrightarrow{y \rightarrow 0} \frac{y^3}{3} + \mathcal{O}(y^5). \quad (22)$$

Thus Eq. (21) has a T^5 low-temperature Debye–Raman asymptote and crosses over toward T^2 when $T \gtrsim \Theta_{\text{eff}}$.

The extracted Θ_{eff} is an effective spectral cutoff of the relevant two-phonon channels and should not be identified with the bulk Debye temperature of diamond.

2.5 Unified DTQEM picture

The three phenomena considered here probe different aspects of the same phonon bath:

Phenomenon	Physical origin	Effective form	Status
$D(T)$	Static shift	$D'_0 + An(\tilde{\nu}, T)$	Fitted in Paper I
$\Gamma_\phi(T)$	Fluctuation ansatz	$\Gamma_{\text{bg}} + \alpha \coth[hc\tilde{\nu}_\phi/(2k_{\text{B}}T)]$	Testable ansatz
$1/T_1(T)$	Raman transitions	$\Gamma_0 + R(T/\Theta_{\text{eff}})^5 J_5(\Theta_{\text{eff}}/T)$	Tested here

Although these expressions share a common Bose–Einstein occupation language, the characteristic energy scales $\tilde{\nu}$, $\tilde{\nu}_\phi$, and Θ_{eff} need not coincide.

3 Experimental Validation

3.1 Datasets and fitting procedure

We test the transport integral model primarily on the Jarmola *et al.* ensemble datasets and discuss Lin *et al.* and Barbosa *et al.* as additional consistency checks. Parameter uncertainties were evaluated using bootstrap resampling with 10^4 iterations. Because the Jarmola data were digitized from published figures, the quoted uncertainties should be regarded as effective fit uncertainties rather than original experimental standard errors.

3.2 Results on Jarmola et al.

Table 1 compares the single-mode Raman ansatz and the J_5 transport model. The single-mode model can achieve a comparable R^2 , but only by pushing the fitted effective frequency above the one-phonon diamond optical phonon scale.

Table 1: Fit results for the Jarmola *et al.* $T_1(T)$ data.

Sample	Model	R^2	Characteristic scale	Physically interp
S2	$n(n+1)$ single mode	0.9954	$\omega \sim 1800 \text{ cm}^{-1}$	No [†]
	J_5 transport	0.9954	$\Theta_{\text{eff}} = 581 \pm 73 \text{ K}$	Yes
S8	$n(n+1)$ single mode	0.9922	$\omega \sim 1800 \text{ cm}^{-1}$	No [†]
	J_5 transport	0.9922	$\Theta_{\text{eff}} = 608 \pm 81 \text{ K}$	Yes
S3	$n(n+1)$ single mode	0.9916	$\omega \sim 1800 \text{ cm}^{-1}$	No [†]
	J_5 transport	0.9916	$\Theta_{\text{eff}} = 599 \pm 94 \text{ K}$	Yes

[†]Exceeds the one-phonon optical phonon scale of diamond ($\approx 1332 \text{ cm}^{-1}$), which is unphysical for a single-m

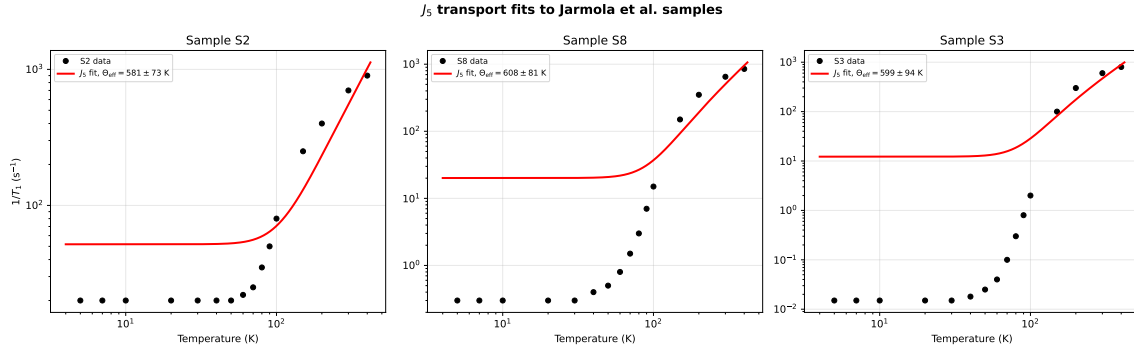


Figure 1: Fit of the J_5 transport integral model to the Jarmola *et al.* $T_1(T)$ data for samples S2, S8, and S3. The Θ_{eff} values shown in the legends are fixed to the Table 1 values. The model captures the overall crossover from a low-temperature plateau to a rapidly increasing high-temperature relaxation rate.

The three fitted central values cluster around

$$\bar{\Theta}_{\text{eff}} = 596 \text{ K}. \quad (23)$$

The sample-to-sample standard deviation of the three central values is approximately 14 K, smaller than the individual bootstrap uncertainties. We therefore quote

$$\Theta_{\text{eff}} = 596 \pm 88 \text{ K}, \quad (24)$$

where the uncertainty represents a typical bootstrap fit uncertainty rather than the sample-to-sample spread.

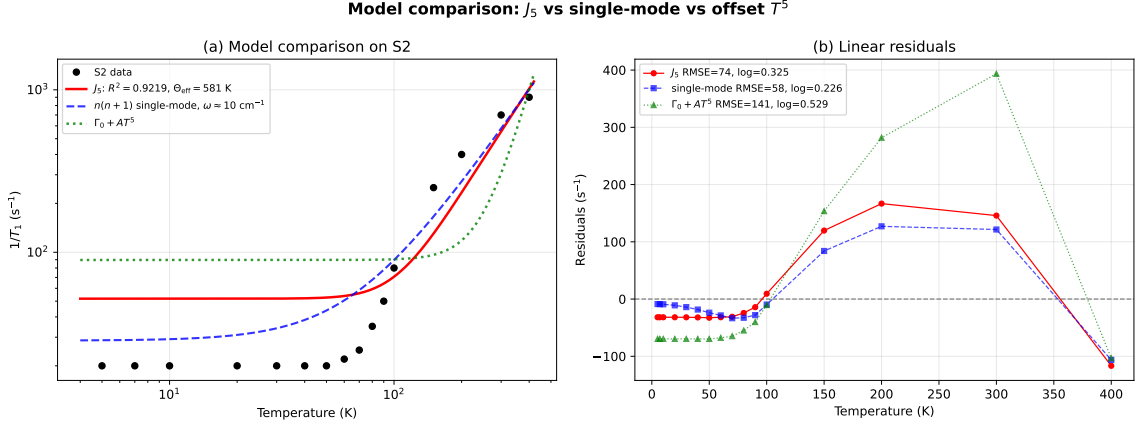


Figure 2: Comparison of three models on the Jarmola S2 dataset: J_5 transport integral, $n(n+1)$ single-mode Raman model, and the offset power law $\Gamma_0 + AT^5$. Linear residuals are shown; logarithmic error metrics are provided in the accompanying code and data files.

3.3 Model comparison on sample S2

4 Discussion

4.1 Why a single mode suffices for $D(T)$ but not for $T_1(T)$

Within the effective description of Paper I, $D(T)$ is represented by a static thermal displacement scale. Such a thermal average can be captured by a single effective phonon energy. In contrast, $T_1(T)$ is a dynamic transition rate. Raman relaxation requires phonon pairs satisfying $|\omega_1 - \omega_2| \approx \omega_s$. A single perfectly sharp mode cannot provide a continuum of near-degenerate phonon pairs with the required frequency differences.

4.2 Why J_5 succeeds physically despite similar R^2

The value of the J_5 model is not primarily that it yields a much larger R^2 . Rather, its advantage is physical interpretability. The preference for J_5 is therefore physical: the single-mode fit reproduces the data only by assigning an effective frequency above the one-phonon diamond optical phonon scale, while J_5 explicitly represents the continuum character of a two-phonon Raman process and yields a physically interpretable continuum cutoff.

4.3 Limitations and outlook

The present framework is intentionally effective. It should be viewed as a compact intermediate description between purely empirical power laws and full *ab initio* spin-phonon theory. Its main limitations are:

1. The J_5 model uses a minimal ω^4 spectral weight.
2. The pure dephasing expression in Eq. (8) remains a testable ansatz.
3. The extracted Θ_{eff} may include effective contributions from Raman, local-mode, and Orbach-like processes over the fitted window.

4. Digitization of published figures introduces uncertainties; fits to raw experimental data would reduce these uncertainties.

5 Conclusion

We have extended the DTQEM phonon-occupation framework from the static zero-field splitting $D(T)$ to spin-lattice relaxation and proposed a corresponding fluctuation ansatz for pure dephasing. The single-mode picture that succeeds for the static shift fails as a physical description of Raman relaxation because the factor $n(n+1)$ has a T^2 high-temperature ceiling within the fixed-frequency single-mode Raman ansatz. Replacing the discrete mode by an effective Debye–Raman transport integral yields a physically interpretable continuum model with $\Theta_{\text{eff}} = 596 \pm 88$ K across three Jarmola ensemble samples. The present work is not a substitute for microscopic spin-phonon theory; it offers a compact, organized bridge between effective models and experimental data, and suggests testable forms for future dephasing experiments.

Data Availability

Digitized datasets and fitting codes are available at DOI: [10.5281/zenodo.20940620](https://doi.org/10.5281/zenodo.20940620).

Acknowledgments

AI-based tools were used for assistance with language editing, code drafting, literature organization, and manuscript preparation. The author takes full responsibility for the scientific content, calculations, and conclusions.

References

- [1] R. Berramdane, *Paper I: A Single-Mode Occupation-Number Model for $D(T)$ in NV Centers*, (unpublished).
- [2] C. L. Degen, F. Reinhard, and P. Cappellaro, *Rev. Mod. Phys.* **89**, 035002 (2017).
- [3] J. F. Barry *et al.*, *Rev. Mod. Phys.* **92**, 015004 (2020).
- [4] A. Jarmola *et al.*, *Phys. Rev. Lett.* **108**, 197601 (2012).
- [5] A. Norambuena *et al.*, *Phys. Rev. B* **97**, 094304 (2018).
- [6] M. W. Doherty *et al.*, *Phys. Rev. B* **90**, 041201(R) (2014).
- [7] M. C. Cambria *et al.*, arXiv:2404.08388v2 (2024).